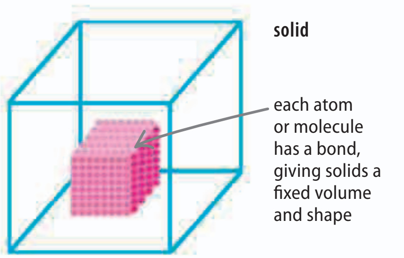
States of matter

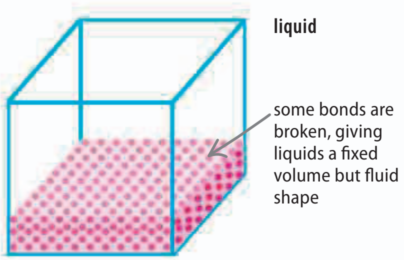
THERE ARE THREE MAIN STATES OF MATTER: SOLID, LIQUID, AND GAS.

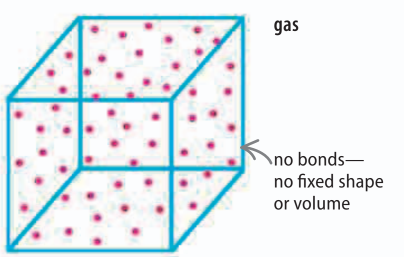
What sets each state apart is how the atoms and molecules (groups of atoms) are bonded together. This bonding is determined by factors such as temperature and pressure.

Physical difference

A solid that is melting into a liquid or boiling into a gas is changing physically. However, all three states share the same chemical formula.



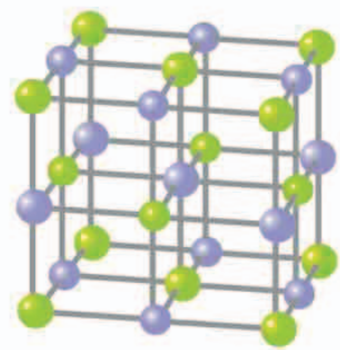




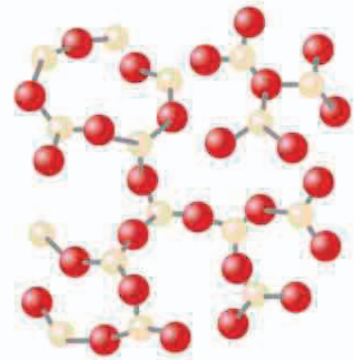
△ **Solid, liquid, gas**
As a substance gets warmer, its molecules break bonds. The substance's structure becomes more chaotic, and changes state, from a solid to a liquid to a gas.

Solids

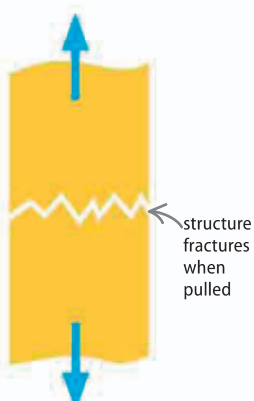
A solid is the most ordered state of matter, with every atom or molecule connected to its neighbors, forming a fixed shape with a fixed volume. Solids are either crystalline, with their units built up in repeating units, or amorphous, with the units grouped together randomly.



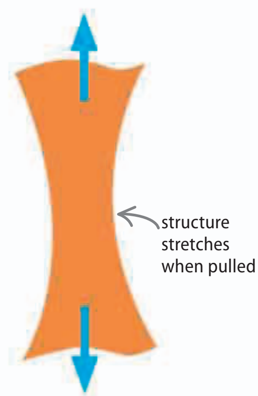
△ **Crystalline halite**
Large crystals of common salt are called halite. The crystal is made up of sodium and chlorine atoms arranged in a cube.



△ **Amorphous silica**
Glass is silica, the same material found in sand. It has an amorphous structure, with the units arranged randomly.



△ **Brittle solid**
In a brittle solid, the particles are held in a crystalline structure. Small forces do not alter the solid, but a force stronger than the bonds between the molecules can break it.



△ **Ductile solid**
Metals and some other solids can be pulled into a wire without breaking. This is because their molecules are held in an amorphous structure and can slide past each other.